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|-----------------------------------|---------------------------------|
| Student Name: <u>James Stuart</u> | <b>Final Examination Part I</b> |
| Date: <b>Monday May-04, 2026</b>  | <b>Time: 01:00pm-03:00pm</b>    |

(30) + (50) = 80

**Question #01 (5pts):** Select the correct answer for the Gibbs free energy  $G^{ID}$  of an ideal liquid mixture of two components with molar composition  $\{x_1, x_2\}$ .

|                                     |                                                                                    |
|-------------------------------------|------------------------------------------------------------------------------------|
| <input type="checkbox"/>            | $G^{ID} = x_1 \bar{G}_1 + x_2 \bar{G}_2$                                           |
| <input type="checkbox"/>            | $G^{ID} = x_1 \bar{G}_1^{ID} + x_2 \bar{G}_2^{ID}$                                 |
| <input checked="" type="checkbox"/> | $G^{ID} = x_1 \bar{G}_1^{ID} + x_2 \bar{G}_2^{ID} + RT(x_1 \ln x_1 + x_2 \ln x_2)$ |

0

**Question #02 (5pts):** Select the correct answer for the definition of the  $K$ -values of an ideal gas mixture in equilibrium with a non-ideal liquid mixture.

✱

|                                     |                                               |
|-------------------------------------|-----------------------------------------------|
| <input type="checkbox"/>            | $K_i = \frac{\hat{\phi}_i^V}{\hat{\phi}_i^L}$ |
| <input type="checkbox"/>            | $K_i = \frac{P_i^{\text{sat}}}{P}$            |
| <input checked="" type="checkbox"/> | $K_i = \hat{\phi}_i^L$                        |

5

**Question #03 (5pts):** Select the correct answer for the equation of dew point temperature of a non-ideal vapor mixture with molar composition  $\{y_1, y_2, \dots, y_N\}$ . Assume ideal-liquid phase mixture

✱

|                                     |                                                              |
|-------------------------------------|--------------------------------------------------------------|
| <input type="checkbox"/>            | $\sum_{i=1}^N y_i \hat{\phi}_i^V = 1$                        |
| <input type="checkbox"/>            | $\sum_{i=1}^N y_i \hat{\phi}_i^L = 1$                        |
| <input checked="" type="checkbox"/> | $\sum_{i=1}^N y_i \frac{\hat{\phi}_i^V}{\hat{\phi}_i^L} = 1$ |
| <input type="checkbox"/>            | $\sum_{i=1}^N y_i \frac{\phi_i^L}{\hat{\phi}_i^V} = 1$       |

5

What is the pressure at the beginning and the end of the vaporization process. Saturated vapor pressures of benzene and toluene are 1008 mmHg and 404 mmHg respectively at 90 °C

$$0 = \sum \frac{z_i (K_i - 1)}{1 + \alpha (K_i - 1)}$$

$$z_i = 0.5 \quad K_i = \frac{P_{sat}}{P}$$

|                                  |     |
|----------------------------------|-----|
| Pressure at the beginning (mmHg) | 706 |
| Pressure at the end (mmHg)       | 288 |

$$\alpha = 0, \text{ sat liquid}$$

$$\alpha = 1 \quad 0 = \sum \frac{z_i (K_i - 1)}{K_i} = \sum z_i - \frac{1}{K_i} = 0$$

$$0 = 0.5 \left( \frac{404}{P} - 1 \right) + 0.5 \left( \frac{1008}{P} - 1 \right)$$

$$0.5 - \frac{P}{404} + 0.5 - \frac{P}{1008} = 0 \rightarrow \frac{P}{404} + \frac{P}{1008} = 1$$

$$\frac{404}{P} + \frac{1008}{P} = 1 \Rightarrow 2P = 1412 \quad P = 706$$

$$P =$$

**Question #05 (5pts):** The following Equation of State  $\frac{PV}{RT} = 1 + \frac{b}{V}$ , where  $b$  is a constant number, describes

the properties of a pure fluid. Derive a formula for the Residual Entropy as a function of  $(Z, B)$  where  $B = \frac{bP}{RT}$

$$\frac{H^R}{RT} = Z - 1 - \int_0^1 T \frac{\partial Z}{\partial T} \frac{d\phi}{\phi}$$

$$\frac{S^R}{R} = \frac{H^R}{RT} - \ln \phi$$

$$B = \frac{bP}{RT}$$

$$\frac{S^R}{R} = \ln Z - \int_0^1 \left[ T \frac{\partial Z}{\partial T} + (Z-1) \right] \frac{d\phi}{\phi}$$

$$Z = 1 + \frac{B}{Z} = 1 + \frac{bP}{RTZ}$$

$$\frac{S^R}{R} = Z - 1 - \int_0^1 T \frac{\partial Z}{\partial T} \frac{d\phi}{\phi} - Z + 1 + \ln Z - \int_0^1 (Z-1) \frac{d\phi}{\phi} = \ln Z - \int_0^1 \left[ T \left( \frac{\partial Z}{\partial T} \right) + (Z-1) \right] \frac{d\phi}{\phi}$$

$$Z^2 - Z = B \rightarrow Z \frac{dZ}{dT} - \frac{dZ}{dT} = - \frac{bP}{RT^2} \quad \frac{dZ}{dT} = - \left( \frac{bP}{RT^2} \right) \frac{1}{Z^2 - 1}$$

**Question #06 (5pts):** The Equilibrium constant of a chemical reaction is independent of the pressure of the reaction mixture.

|                                     |       |
|-------------------------------------|-------|
| <input checked="" type="checkbox"/> | True  |
| <input type="checkbox"/>            | False |

**Question #07 (5pts):** An ideal gas expands **isenthalpically** from a pressure  $P_1 = 10 \text{ bar}$  to a pressure  $P_2 = 1 \text{ bar}$ . What is the temperature change of the gas during this process (select from the list below)

|                                     |                |
|-------------------------------------|----------------|
| <input checked="" type="checkbox"/> | $\Delta T = 0$ |
| <input type="checkbox"/>            | $\Delta T < 0$ |
| <input type="checkbox"/>            | $\Delta T > 0$ |

**Question #08 (5pts):** Consider the total differential  $d(nG) = nVdP - nSdT + \sum \bar{G}_i dn_i$ . Which formulas from the ones shown below are correct. You must select all that apply for full credit

$$d(nG) = nVdP - nSdT + \sum \bar{G}_i dn_i$$

|                                     |                                                                                                             |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> | $\left( \frac{\partial S}{\partial P} \right)_{T,n} = - \left( \frac{\partial V}{\partial T} \right)_{P,n}$ |
| <input checked="" type="checkbox"/> | $\left( \frac{\partial \bar{G}_i}{\partial P} \right)_{T,n_j, n_i} = \bar{V}_i$                             |
| <input checked="" type="checkbox"/> | $\left( \frac{\partial \bar{G}_i}{\partial T} \right)_{P,n_j, n_i} = -\bar{S}_i$                            |

Question #09 (5pts): The fugacity coefficient of a component inside an ideal-liquid mixture is equal to one.

|                                     |       |
|-------------------------------------|-------|
| <input checked="" type="checkbox"/> | True  |
| <input type="checkbox"/>            | False |

0

Question #10 (5pts): A single VLE experimental data point is available at the bubble point temperature  $T_{BUBBLE} = 35^\circ\text{C}$  under pressure  $P = 76.719\text{ kPa}$  with composition  $[x_1, y_1] = [0.1665, 0.0832]$  for a binary mixture. The saturated vapor pressures of the two components are  $P_1^{\text{sat}} = 49.624\text{ kPa}$  and  $P_2^{\text{sat}} = 85.265\text{ kPa}$  at the bubble point temperature. What is the value of the coefficient  $A$  in the Margules 2-suffix equation for the Excess Gibbs Free Energy. Assume ideal gas mixture for the vapor phase.

Value of  $A$  | ~~-1.8796~~

2

$$\ln \gamma_i = A x_1 x_2$$

$$y_i = \frac{\hat{f}_i}{x_i \gamma_i}$$

$$K_i = y_i \frac{P^{\text{sat}}}{P} = \frac{y_i}{x_i} \quad f^v = P$$

$$\gamma_i = \frac{y_i}{x_i} \frac{P}{P^{\text{sat}}}$$

$$[x_2, y_2] = [0.8335, 0.9168]$$

$$\gamma_1 = \frac{0.0832}{0.1665} \frac{76.719}{49.624} = 0.77254$$

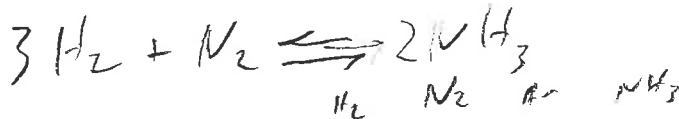
$$\ln(0.77254) = A(0.1665)(0.8335)$$



**UNIVERSITY OF ILLINOIS CHICAGO**  
**Department of Chemical Engineering**  
**CHE 301: Chemical Engineering Thermodynamics**

|                                   |                                  |
|-----------------------------------|----------------------------------|
| Student Name: <u>James Stuart</u> | <b>Final Examination Part II</b> |
| Date: <b>Monday May-04, 2026</b>  | Time: <b>01:00pm-03:00pm</b>     |

**Problem #01 (15 pts):** A gas mixture containing 70 mol% Hydrogen, 20 mol% Nitrogen and 10 mol% inert gas Argon is passed over a catalyst for production of Ammonia. If the pressure of the reaction mixture is 50 atm and the final temperature 400 °C, calculate the percentage of Hydrogen converted to Ammonia and the mole fraction of Ammonia in the reaction mixture. Assume the gas mixture follows the Peng-Robinson Equation of State with Binary Interaction Parameters all set equal to zero.



|                               |        |
|-------------------------------|--------|
| H <sub>2</sub> Conversion (%) | 23.96% |
| NH <sub>3</sub> mole fraction | 0.1259 |

15

Initial mols: 70, 20, 10, 0

Guss extent = 10

Input T and P, calculate z and φ

Calculate  $K_{eq}$  Gibbs and  $K_{eq}$ , Newton method to make  $K_{eq} = K_{eq}$  Gibbs by changing extent

H<sub>2</sub> conv. % =  $\frac{\text{Initial H}_2 - \text{Final H}_2}{\text{Initial H}_2} \times 100$

**Problem #02 (15 pts):** The following Equation of State has been proposed for the calculation of the thermal properties of Nitrogen:

$$P + \frac{a}{TV^2} = \frac{RT}{V-b} \quad \left| \quad a = \lambda_1 \frac{R^2 T_c^3}{P_c} \quad \right| \quad b = \lambda_2 \frac{RT_c}{P_c}$$

Calculate the values of the numerical constants  $\{\lambda_1, \lambda_2\}$  and the value of the critical compressibility factor  $Z_c$ . Derive an expression for the calculation of the residual enthalpy and calculate its value at  $T=50^\circ\text{C}$  and pressure  $P=650\text{ atm}$ . **Hint:** Use an initial guess of  $Z^{(0)} = 2$  for the calculation of the residual enthalpy

8.314  $\frac{\text{J}}{\text{mol K}} = 0.08314 \frac{\text{bar m}^3}{\text{mol K}}$

$\frac{\text{bar}^2 \text{m}^6}{\text{mol}^2 \text{K}^2} \cdot \text{K}^3$

$\frac{\text{bar}^2 \text{m}^6}{\text{mol}^2 \text{K}^2} \cdot \text{K}$

|                  |        |
|------------------|--------|
| $\lambda_1$      | 226.23 |
| $\lambda_2$      | 96.66  |
| $Z_c$            | 0.375  |
| $\frac{H^R}{RT}$ |        |

eq 1:  $P = \frac{RT}{V-b} - \frac{a}{TV^2}$

eq 2:  $\frac{RT}{(V-b)^2} = \frac{2a}{TV^3}$  First derivative = 0

eq 3:  $\frac{2RT}{(V-b)^3} = \frac{6a}{TV^4}$  Second derivative = 0

$\frac{6(3)}{(2)} \div \frac{2}{V-b} = \frac{3}{V} \rightarrow b = \frac{1}{3} V$

$a = \frac{RT^2 V^3}{2(V-b)^2}$

$\lambda_1 = \frac{6 P_c}{RT_c}$

$\lambda_2 = \frac{9 P_c}{R^2 T_c^3}$

$\frac{PV}{RT} = \left[ \frac{RT}{Vb} - \frac{a}{TV^2} \right] \frac{V}{RT}$

$Z_c = \text{this eqn at critical P}$

$\frac{H^R}{RT} = Z - 1 - \frac{3a}{2RT^2 V}$

**Problem #03 (15 pts):** The mixture of Methanol (Component #1), Acetone (Component #2) and Cyclohexane (Component #3) at  $T = 101^\circ\text{C}$  forms an azeotrope  $\{y_1 = x_1, y_2 = x_2 \text{ and } y_3 = x_3\}$ . Determine the pressure and composition that is predicted for the azeotrope between methanol, acetone and cyclohexane by using the NRTL method with binary interaction parameters,  $\{A_{ij}, A_{ji}, B_{ij}, B_{ji}, C\}$  shown in the table and assuming that the vapor phase follows the **Redlich-Kwong** Equation of State with all BIPs set equal to zero. **Poynting Factor=1**

| BIPs                   |  | CH <sub>4</sub> O<br>C <sub>3</sub> H <sub>6</sub> O | CH <sub>4</sub> O<br>C <sub>6</sub> H <sub>12</sub> | C <sub>3</sub> H <sub>6</sub> O<br>C <sub>6</sub> H <sub>12</sub> |
|------------------------|--|------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------|
| <b>K<sub>Aij</sub></b> |  | 0.0000                                               | 0.0000                                              | 0.0000                                                            |
| <b>K<sub>Bij</sub></b> |  |                                                      |                                                     |                                                                   |
| <b>K<sub>Cij</sub></b> |  |                                                      |                                                     |                                                                   |
| <b>A<sub>ij</sub></b>  |  | 0.0000                                               | 1.3869                                              | -0.1061                                                           |
| <b>A<sub>ji</sub></b>  |  | 0.0000                                               | -4.6753                                             | -3.5142                                                           |
| <b>B<sub>ij</sub></b>  |  | 114.1347                                             | 224.7601                                            | 359.6793                                                          |
| <b>B<sub>ji</sub></b>  |  | 101.8859                                             | 2277.7942                                           | 1485.3369                                                         |
| <b>C</b>               |  | 0.3000                                               | 0.4300                                              | 0.4700                                                            |

|                           |        |
|---------------------------|--------|
| Pressure (bar)            | 4.6165 |
| Methanol mole fraction    | 0.5899 |
| Acetone mole fraction     | 0.1347 |
| Cyclohexane mole fraction | 0.2754 |

Guess  $z_i = [0.4, 0.3, 1 - z_1 - z_2]$

$x_i = z_i$ , Guess  $P_{\text{BUB}} = \text{Highest } P_{\text{sat}}$

Calc  $\gamma_i$

Calc  $P_i^{\text{sat}}$ ,  $K_i = \gamma_i \frac{P_i^{\text{sat}}}{P}$

Goal seek  $RR_{\text{yca}} = 0$

$y_i = x_i K_i$

Solver,  $RR=0$ ,  $K_1 = K_2 = 1$ , manipulation  $P_{\text{BUB}}, z_1, z_2$

**Problem #04 (15 pts):** A liquid hydrocarbon mixture of Methane (Comp #1), Ethane (Comp #2) and Propane (Comp #3) at temperature  $T_1 = -100^\circ\text{C}$  and pressure  $P_1 = 64\text{ bar}$  with molar composition  $z = [0.80, 0.08, 0.12]$  expands **adiabatically** down to a pressure of  $P_2 = 10\text{ bar}$  in a liquid turbine. If the temperature at the outlet of the liquid turbine is  $T_2 = -118.5^\circ\text{C}$  calculate the power generated in kW. The flow rate of the hydrocarbon mixture is 10,000 kmol/hr. Also report the calculated vapor fraction at the turbine outlet and the Lost Work in kW. Assume that both phases of the hydrocarbon mixture follow the Peng-Robinson Equation of State with all BIPs set equal to zero. Take the temperature of the infinite environment to be equal to  $T_0 = 300\text{ K}$

|                                      |         |
|--------------------------------------|---------|
| Power (kW)                           | 784.5   |
| Vapor fraction at the turbine outlet | 0.1313  |
| Lost Work (kW)                       | 38.7872 |

15

Calc bubble point  $P = 10.3\text{ bar}$

Do PT flash calc:

Must modify  $P_{\text{sat}}$ , multiply by 6.1

PT flash calc shows  $Q = 0.1313$

Do energy balance, lever rule for outlet

$$W = H_2 - H_1$$

$$B = H - 300\text{ S}$$

$$LW = B_1 - B_2 + W$$